C: Separation properties

- Again we start with a baby theorem:

Baby-Theorem

Suppose $a < b < c < d$. Then one can find a continuous function $f : \mathbb{R} \to \mathbb{R}$ s.t. $f([a, b]) = 0, f([c, d]) = 1$.

- It is not hard to extend this baby theorem to

Theorem (Urysohn Lemma in $\mathbb{R}^n$)

Suppose $K, L \subset \mathbb{R}^n$ are closed sets, and $K \cap L = \emptyset$. Then $\exists$ continuous function $f : \mathbb{R}^n \to \mathbb{R}$ s.t. $f(K) = 0, f(L) = 1$.

C: Separation properties

- Urysohn's lemma is useful because it is equivalent to the following extension theorem:

Theorem (Tietze Extension Theorem)

Suppose $K \subset \mathbb{R}^n$ is closed, f is a continuous function defined on K. Then f can be extended to a continuous function on $\mathbb{R}^n$.

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- Topological property behind these two Theorems:

separation properties of $\mathbb{R}^n$.

C: Separation properties

Again we can extend these two theorems to more general topological spaces. For example, we will prove

Theorem (Tietze extension theorem)

Suppose X is a normal topological space, and $A \subset X$ is closed. Then any continuous function $f : A \to \mathbb{R}$ can be extended to a continuous function $F : X \to \mathbb{R}$.

C: Separation properties

Both compactness and the separation property will be needed in proving the famous Stone-Weierstrass theorem (in a general setting):

Theorem (Stone-Weierstrass)

Let X be any compact Hausdorff space, and $L \subset C(X, \mathbb{R})$ be a sub-algebra which vanishes at no point and separate points. Then L is dense.

Part II: Algebraic/Geometric Topology

D: Brouwer Fixed Point Theorem

Again we start with a baby theorem that we learned in mathematical analysis:

Baby-Theorem (Baby fixed point theorem)

Let $f : [a, b] \to [a, b]$ be any continuous function. Then $\exists c \in [a, b]$ s.t. $f(c) = c$.

D: Brouwer Fixed Point Theorem

In this course we will prove

Theorem (Brouwer Fixed Point Theorem for $n = 2$)

Let $\mathbb{D}$ be a planar disc and $f : \mathbb{D} \to \mathbb{D}$ be any continuous map. Then $\exists p \in \mathbb{D}$ s.t. $f(p) = p$.





D: Brouwer Fixed Point Theorem

Some thoughts:

- Obviously the fixed point theorem fails if you replace $[a, b]$ by $(a, b]$ or $[a, b] \cup [c, d]$, which indicates the need of compactness and connectedness.
- It also fails if you replace $\mathbb{D}$ by a closed planar annulus. The reason is that you can find a circle in the annulus which cannot be deformed to a point. This is a *higher level connectedness*, which is described by the theory of *fundamental groups* (and more generally, *homotopy groups*).

D: Borsuk-Ulam theorem

After developing the theory of fundamental groups, we will prove the 2-dimensional Brouwer fixed point theorem as well as the Borsuk-Ulam theorem for $n = 2$:

Theorem (Borsuk-Ulam Theorem)

There does not exists any antipodal continuous map $f : S^n \to S^{n-1}$.

D: The Ham-Sandwich Theorem

As a consequence of Borsuk-Ulam theorem ($n = 3$), one has

Theorem (The Ham-Sandwich Theorem)

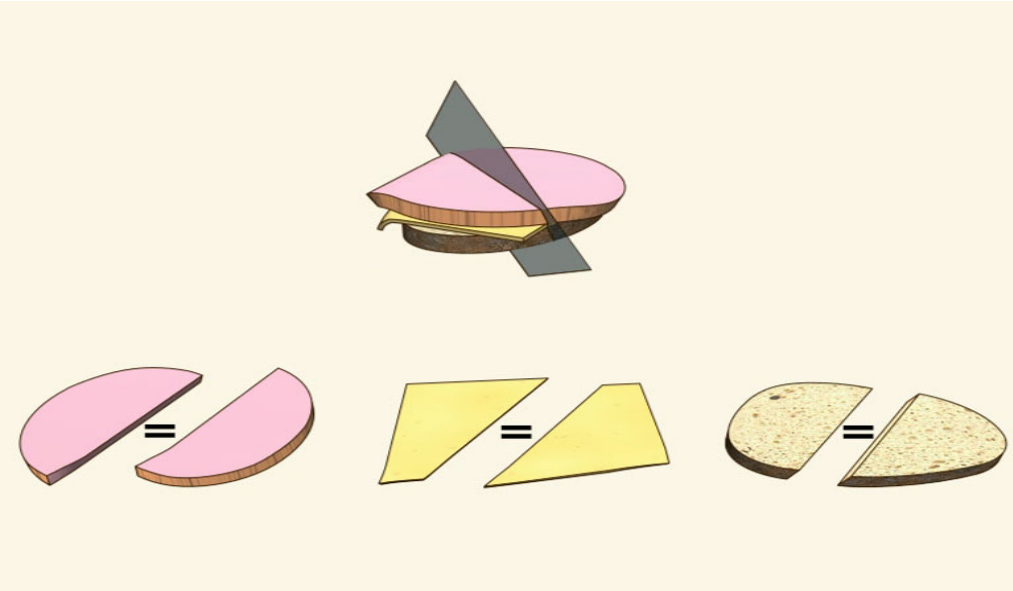
Given any sandwich consisting of bread, cheese and ham, one can cut the sandwich into two pieces by a single knife slice so that each piece has exactly half bread, half cheese and half ham.

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D: Brouwer Fixed Point Theorem

Again these theorems generalize to higher dimension. However, the proofs are more involved: the fundamental groups are not enough. We will not try to develop the general theory of homotopy groups or homology groups to prove these theorems. However, we will prove the general Brouwer fixed point theorem via ideas from *differential topology*:

Theorem (Brouwer Fixed Point Theorem for arbitrary n)

Let $\mathbb{B} \subset \mathbb{R}^n$ be a ball and $f : \mathbb{B} \to \mathbb{B}$ be any continuous map. Then $\exists p \in \mathbb{B}$ s.t. $f(p) = p$.

D: Brouwer Invariance of domain

With some hard work, we can prove

Theorem (Brouwer's Invariance of domain)

If $U \subset \mathbb{R}^n$ is open, $f : U \to \mathbb{R}^n$ is injective and continuous, then $f(U)$ is open in $\mathbb{R}^n$.

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which implies the following theorem that looks "obviously true":

Theorem (Topological of dimension)

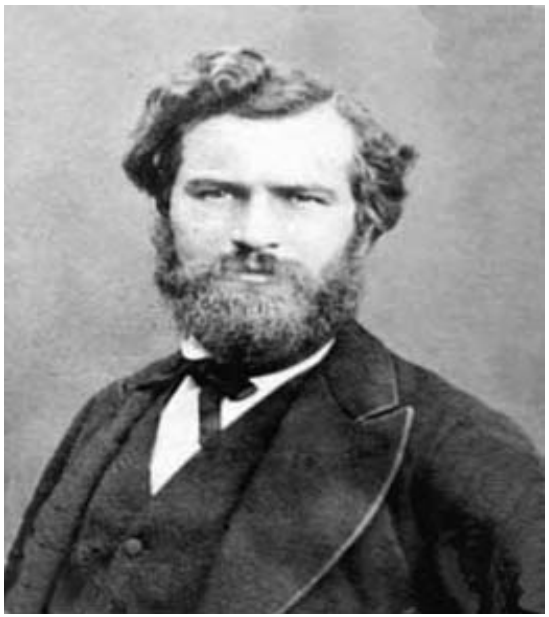
If $n \neq m$, then (in topology) $\mathbb{R}^n \neq \mathbb{R}^m$.

E: Jordan curve theorem

Here is another theorem which "is obviously true" but the proof is very complicated:

Theorem (Jordan Curve Theorem)

Any simple closed curve in the plane separate the plane into two disjoint regions.

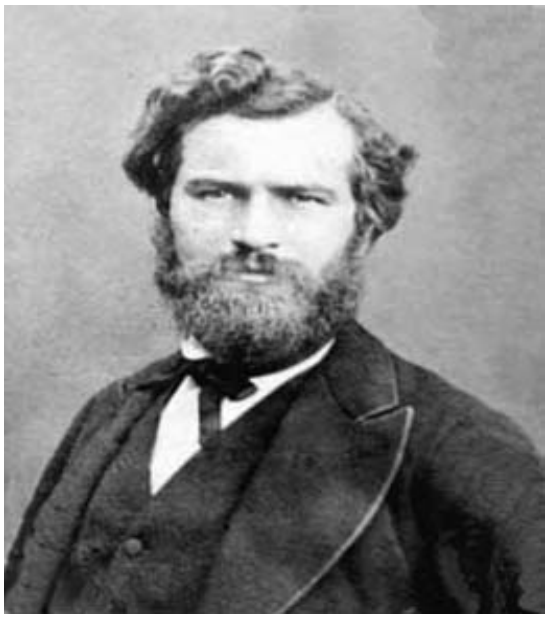


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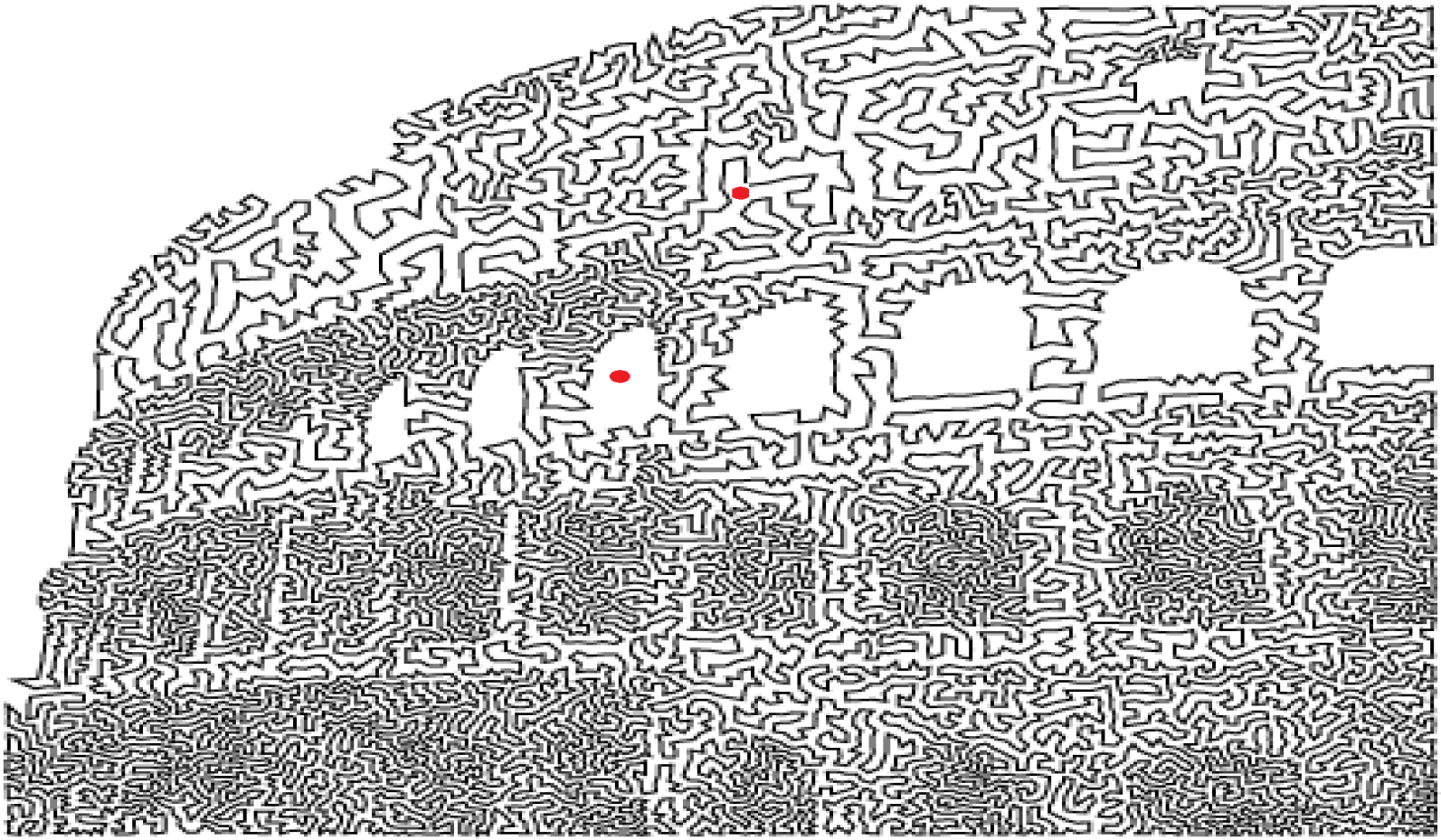


Camille Jordan (1838-1922) was also known for:

- Jordan normal form (linear algebra)
- Jordan measure (real analysis)

E: Jordan curve theorem

A simple closed curve need not be very simple:



E: Jordan-Brouwer

Similar (but weaker) theorem holds in higher dimension, which we will not prove in this course:

Theorem (Jordan-Brouwer separation theorem)

If X is a subset in $\mathbb{R}^n$ which is homeomorphic to S^{n-1}, then $\mathbb{R}^n \setminus X$ has exactly two components, one is bounded and the other is unbounded, so that X is their common boundary.

F: Euler Polyhedron Formula

Finally let's return to Euler's work. In 1750 Euler proved one of his most famous formula[9]:

Theorem (Euler Polyhedron Formula, 1750)

For any convex polyhedron,

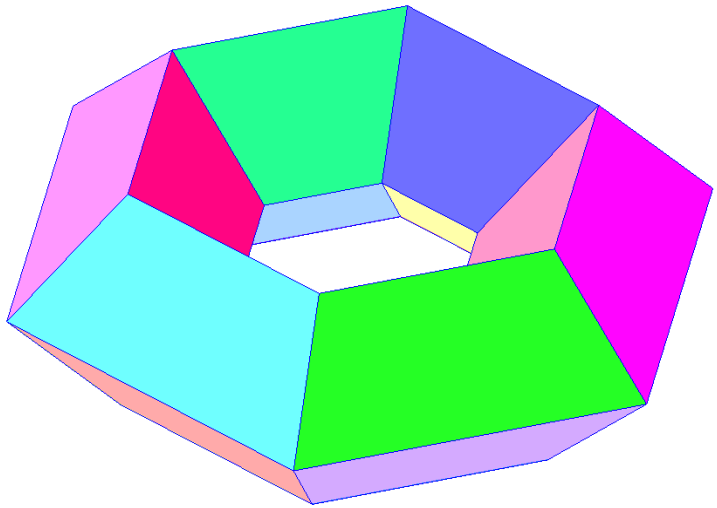
$$V - E + F = 2,$$

where V, E, F denotes the number of vertices, edges and faces respectively.

[9] Some people regard this theorem as the birth of topology.

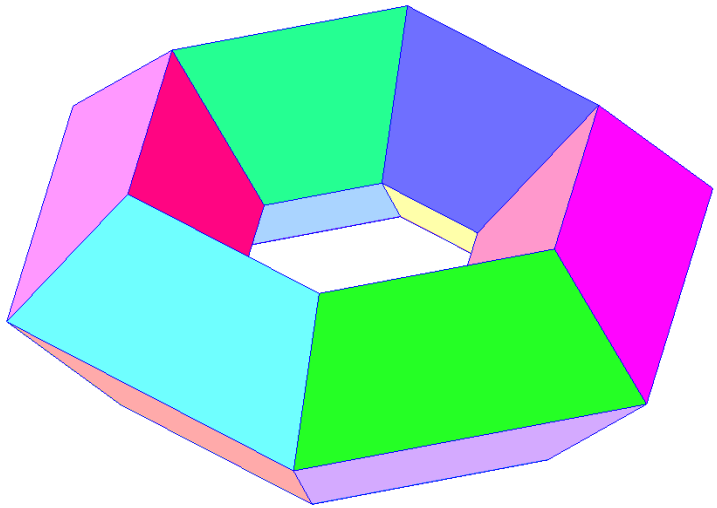
F: Euler Polyhedron Formula

You may ask: what about other polyhedrons? For example,



F: Euler Polyhedron Formula

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If you count the number of vertices, edges and faces of this torus-like polyhedron, you will find

$$V - E + F = 24 - 48 + 24 = 0.$$

F: Euler Polyhedron Formula

It turns out that the numbers 2 and 0 are topological:

- 2 is the Euler characteristic of S^2 (which is homeomorphic to convex polyhedrons in $\mathbb{R}^3$)
- 0 is the Euler characteristic of the torus T^2.

F: Euler Polyhedron Formula

In general, one has

Theorem (Generalized Euler Polyhedron Formula, L'Huilier 1812)

If polyhedron $P \simeq$ surface S, then

$$V - E + F = \chi(S).$$

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If polyhedron $P \simeq$ surface S, then

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Here, the Euler characteristic $\chi(S)$ of an (oriented) closed surface S is defined to be

$$\chi(S) = 2 - 2k,$$

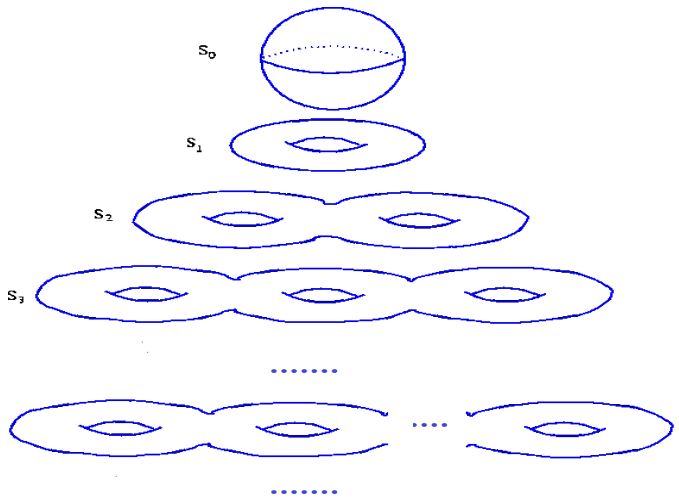
where k equals "the number of holes" of S.

F: Classification of surfaces

We will prove the following classification theorem of surfaces[10]:

Theorem (Classification theorem of oriented compact surfaces)

The list of all oriented compact surfaces without boundary:



[10]We will prove a more general version: the classification of compact surfaces (including non-orientable surfaces and compact surfaces with boundary).

F: Poincaré-Hopf Theorem in 2-dim

As an application, we can prove

Theorem (Poincaré-Hopf)

Let S be a connected compact oriented surface without boundary, and V a continuous vector field on S with only isolated critical points. Then

$$\sum_p \mathrm{ind}_p V = \chi(S).$$



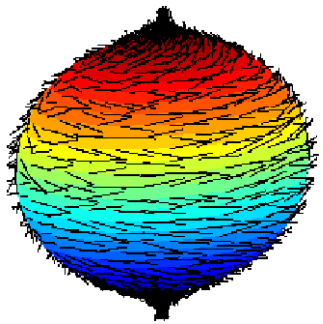


F: Hairy ball theorem

Since $\chi(S^2) = 2 \neq 0$, we get

Corollary (Hairy ball theorem)

Any continuous vector field on the sphere S^2 must vanish at some point.





Two sayings

Let's end this lengthy introduction by two sayings:

Mathematicians do not study objects, but relations between objects.

– H. Poincare

"Obvious" is the most dangerous word in mathematics.

– E. Bell